

Big Data Analytics and Data Science Fundamentals

Big Data Analytics

Definition

- Examining large complex datasets
- Uncovering hidden patterns
- Extracting meaningful information

The 4 V's

- Volume: Scale of data
- Velocity: Speed of processing
- Variety: Diverse data types
- Veracity: Data trustworthiness

Process Workflow

- Collect: Structured and unstructured
- Process: Batch vs Stream
- Clean: Scrubbing and formatting
- Analyze: Mining and predictive

Business Importance

- Enhanced Decision-Making
- Operational Efficiency
- Personalized Customer Experience
- Innovation and New Revenue

Challenges

- Data Quality and Integration
- Data Privacy and Security
- Skill Gap
- Implementation Cost

Tools and Tech

- Batch: Hadoop, Spark, Tez
- Streaming: Kafka, Storm, Flink
- Cloud: Kinesis, Dataflow

Data Science

Definition

- Multidisciplinary field
- Scientific methods and algorithms
- Extracting knowledge from data

Process Stages

- Problem Definition
- Data Collection
- Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Model Building and Evaluation
- Deployment and Maintenance

Industry Applications

- Healthcare: Disease prediction
- Finance: Fraud detection
- Retail: Recommendation engines
- Transportation: Route optimization
- Manufacturing: Predictive maintenance

Comparison

Traditional Processing

- Structured data
- Relational databases
- Limited scalability

Big Data Analytics

- Unstructured data
- Distributed storage
- High scalability